

GrounUp Enterprise Specification

Phase 20 - Enterprise Integration, Deployment, Testing, Governance & Production Readiness

Version 1.0.0 / Review-ready / Proposed lock: GES-P20-v1.0.0

1. Mission

Create the governed enterprise convergence layer that turns the completed GrounUp phase specifications into a deployable, testable, supportable, secure, scalable and recoverable SaaS platform.

2. Architecture Principles

API/event contracts; environment separation; immutable versioned artifacts; infrastructure/configuration as code; zero bypass of release gates; tenant isolation; observable/recoverable services; end-to-end traceability; human-governed AI.

3. Integration Architecture

All internal and external integrations use governed contracts, ownership, authentication, schema/version policies, idempotency/retry behavior, reconciliation and operational health monitoring.

4. Deployment & Release Engineering

Builds produce traceable signed artifacts promoted through controlled environments. Production releases require approved change/release records, test evidence, migration readiness, rollback and post-deployment validation.

5. Testing & Traceability

The platform supports unit through UAT, performance, security, accessibility and DR testing. Phase 00-20 requirements are mapped to tests and implementation surfaces; unresolved blocking gaps prevent enterprise lock.

6. Security, Privacy & Compliance

Threat modeling, vulnerability controls, encryption, identity, tenant isolation, privacy/data controls, evidence collection and access reviews are production-readiness gates.

7. Reliability, Performance & DR

Services define SLIs/SLOs, error budgets, capacity expectations, resilience patterns, backups, restore validation, RTO/RPO and tested failover/recovery procedures.

8. SaaS & White-Label Operations

Tenant provisioning controls plans, entitlements, limits, branding/white-label profiles and lifecycle while maintaining strict isolation and auditable configuration.

9. AI Production Governance

AI assets require versioned models/prompts, evaluations, guardrails, evidence, monitoring, human-review policy, incident handling and rollback. AI cannot bypass platform safety/security/readiness controls.

10. Go-Live, Hypercare & Enterprise Lock

Production readiness culminates in an explicit go/no-go decision, controlled cutover, hypercare, stabilization, operational handoff and final GES v1.0 enterprise baseline after Phase 00-20 reconciliation.

11. Functional Domains

Enterprise Integration Architecture

Canonical integration patterns, contracts, adapters, event flows, orchestration, anti-corruption layers and ownership.

API Governance

API lifecycle, versioning, schemas, authentication, authorization, quotas, documentation, compatibility and deprecation.

Event & Messaging Platform

Events, topics, queues, delivery semantics, ordering, idempotency, retries, dead letters and replay.

Data Integration & Synchronization

Master/reference data exchange, CDC, batch/stream sync, reconciliation, conflict resolution and lineage.

Identity & Access Integration

SSO, federation, SCIM, service identities, RBAC/ABAC, least privilege and tenant boundaries.

External Connectors

ERP, accounting, CRM, HRIS, fleet, telematics, GIS, document, payment, weather, vendor and public-system adapters.

Environment Strategy

Development, test, QA, staging, training, sandbox, pre-production, production and disaster-recovery environments.

Configuration Management

Environment configuration, feature flags, tenant configuration, secrets references, drift control and promotion.

CI/CD & Release Engineering

Source control gates, build, test, package, artifact signing, deployment, promotion, rollback and release evidence.

Infrastructure as Code

Repeatable cloud/infrastructure definitions, modules, state, review, policy checks, drift and recovery.

Container & Runtime Platform

Application packaging, orchestration, health checks, scaling, runtime policy and workload isolation.

Database & Storage Readiness

Schema migration, indexing, partitioning, backups, retention, replication, restore and capacity.

Security Engineering

Threat modeling, secure defaults, vulnerability management, dependency scanning, encryption and security testing.

Privacy & Data Protection

Data classification, minimization, retention, deletion, consent/configuration, export and tenant isolation.

Observability

Logs, metrics, traces, dashboards, alerting, correlation IDs, audit observability and operational telemetry.

Reliability Engineering

SLIs, SLOs, error budgets, resilience, retries, circuit breakers, graceful degradation and failure domains.

Performance & Scalability

Load, stress, soak, concurrency, latency, throughput, capacity, autoscaling and bottleneck analysis.

Disaster Recovery & Business Continuity

RTO/RPO, backups, replication, failover, recovery exercises, runbooks and business continuity.

Testing Governance

Unit, component, integration, contract, system, regression, UAT, performance, security, accessibility and recovery testing.

Automated Test Platform

Test data, fixtures, environments, mocks, service virtualization, suites, execution, evidence and flaky-test controls.

Requirements Traceability & Coverage

Phase 00-20 requirement coverage, test mapping, implementation mapping, exceptions and closure.

Data Migration & Cutover

Source assessment, mapping, cleansing, rehearsal, migration, reconciliation, delta load, cutover and rollback.

Production Readiness Review

Architecture, security, operations, support, data, integrations, performance, DR, training and business readiness gates.

Release & Change Management

Release trains, change records, approvals, maintenance windows, communications, rollback and post-release validation.

Incident & Problem Management

Severity, triage, response, escalation, communication, RCA, corrective actions and problem records.

Service Management & Support

Service catalog, support tiers, SLAs, requests, knowledge, escalation, ownership and customer/tenant support.

Tenant Provisioning & SaaS Operations

Tenant creation, plans, entitlements, branding/white-label configuration, limits, isolation and lifecycle.

Compliance & Audit Readiness

Control evidence, audit trails, policy attestations, access reviews, change evidence and compliance mappings.

AI/ML Production Governance

Model/prompt/version registry, evaluations, guardrails, monitoring, human review, rollback and AI incident response.

Enterprise Go-Live & Hypercare

Cutover command center, launch checklist, monitoring, support staffing, issue burn-down, stabilization and handoff.

Platform Digital Twin & Control Tower

Real-time environment, release, integration, health, risk, readiness, incident and governance state.

12. Controlled Artifact Summary

Artifact	Count
Requirements	450
Entities	210
Fields	350
Business Rules	90
Validation Rules	90
Workflows	60
Engines	66
Api Resources	84
Ui Surfaces	50
Reports	52
Kpis	65
Tests	450

13. KPI Registry

P20-KPI-001 - Integration Availability: Governed metric definition for Integration Availability.

P20-KPI-002 - Connector Success Rate: Governed metric definition for Connector Success Rate.

P20-KPI-003 - API Error Rate: Governed metric definition for API Error Rate.

P20-KPI-004 - API Latency: Governed metric definition for API Latency.

P20-KPI-005 - Event Delivery Success: Governed metric definition for Event Delivery Success.

P20-KPI-006 - Dead Letter Rate: Governed metric definition for Dead Letter Rate.

P20-KPI-007 - Sync Success Rate: Governed metric definition for Sync Success Rate.

P20-KPI-008 - Reconciliation Exception Rate: Governed metric definition for Reconciliation Exception Rate.

P20-KPI-009 - SSO Success Rate: Governed metric definition for SSO Success Rate.

P20-KPI-010 - Environment Drift Count: Governed metric definition for Environment Drift Count.

P20-KPI-011 - Configuration Drift Count: Governed metric definition for Configuration Drift Count.

P20-KPI-012 - Release Success Rate: Governed metric definition for Release Success Rate.

P20-KPI-013 - Deployment Frequency: Governed metric definition for Deployment Frequency.

P20-KPI-014 - Deployment Failure Rate: Governed metric definition for Deployment Failure Rate.

P20-KPI-015 - Rollback Rate: Governed metric definition for Rollback Rate.

P20-KPI-016 - Lead Time for Change: Governed metric definition for Lead Time for Change.

P20-KPI-017 - Mean Time to Restore: Governed metric definition for Mean Time to Restore.

P20-KPI-018 - Change Failure Rate: Governed metric definition for Change Failure Rate.

P20-KPI-019 - Infrastructure Policy Pass Rate: Governed metric definition for Infrastructure Policy Pass Rate.

P20-KPI-020 - Schema Migration Success: Governed metric definition for Schema Migration Success.

P20-KPI-021 - Backup Success Rate: Governed metric definition for Backup Success Rate.

P20-KPI-022 - Restore Test Pass Rate: Governed metric definition for Restore Test Pass Rate.

P20-KPI-023 - Critical Vulnerability Count: Governed metric definition for Critical Vulnerability Count.

P20-KPI-024 - Vulnerability Remediation Time: Governed metric definition for Vulnerability Remediation Time.

P20-KPI-025 - Tenant Isolation Pass Rate: Governed metric definition for Tenant Isolation Pass Rate.

P20-KPI-026 - Observability Coverage: Governed metric definition for Observability Coverage.

P20-KPI-027 - SLO Compliance: Governed metric definition for SLO Compliance.

P20-KPI-028 - Error Budget Remaining: Governed metric definition for Error Budget Remaining.

P20-KPI-029 - Alert Noise Rate: Governed metric definition for Alert Noise Rate.

P20-KPI-030 - Incident MTTD: Governed metric definition for Incident MTTD.

P20-KPI-031 - Incident MTTA: Governed metric definition for Incident MTTA.

P20-KPI-032 - Incident MTTR: Governed metric definition for Incident MTTR.

P20-KPI-033 - Performance P95 Latency: Governed metric definition for Performance P95 Latency.

P20-KPI-034 - Peak Throughput: Governed metric definition for Peak Throughput.

P20-KPI-035 - Capacity Headroom: Governed metric definition for Capacity Headroom.

P20-KPI-036 - Autoscaling Success: Governed metric definition for Autoscaling Success.

P20-KPI-037 - DR Exercise Pass Rate: Governed metric definition for DR Exercise Pass Rate.

P20-KPI-038 - RTO Compliance: Governed metric definition for RTO Compliance.

P20-KPI-039 - RPO Compliance: Governed metric definition for RPO Compliance.

P20-KPI-040 - Automated Test Pass Rate: Governed metric definition for Automated Test Pass Rate.

P20-KPI-041 - Regression Coverage: Governed metric definition for Regression Coverage.

P20-KPI-042 - Requirements Test Coverage: Governed metric definition for Requirements Test Coverage.

P20-KPI-043 - Traceability Coverage: Governed metric definition for Traceability Coverage.

P20-KPI-044 - Flaky Test Rate: Governed metric definition for Flaky Test Rate.

P20-KPI-045 - Open Critical Defects: Governed metric definition for Open Critical Defects.

P20-KPI-046 - UAT Pass Rate: Governed metric definition for UAT Pass Rate.

P20-KPI-047 - Migration Reconciliation Accuracy: Governed metric definition for Migration Reconciliation Accuracy.

P20-KPI-048 - Cutover Task Completion: Governed metric definition for Cutover Task Completion.

P20-KPI-049 - Production Readiness Gate Pass: Governed metric definition for Production Readiness Gate Pass.

P20-KPI-050 - Go-Live Blocker Count: Governed metric definition for Go-Live Blocker Count.

P20-KPI-051 - Service Level Compliance: Governed metric definition for Service Level Compliance.

P20-KPI-052 - Tenant Provisioning Time: Governed metric definition for Tenant Provisioning Time.

P20-KPI-053 - White-Label Configuration Success: Governed metric definition for White-Label Configuration Success.

P20-KPI-054 - Compliance Evidence Coverage: Governed metric definition for Compliance Evidence Coverage.

P20-KPI-055 - Access Review Completion: Governed metric definition for Access Review Completion.

P20-KPI-056 - AI Evaluation Pass Rate: Governed metric definition for AI Evaluation Pass Rate.

P20-KPI-057 - AI Guardrail Violation Rate: Governed metric definition for AI Guardrail Violation Rate.

P20-KPI-058 - AI Production Incident Count: Governed metric definition for AI Production Incident Count.

P20-KPI-059 - Hypercare Issue Burn-Down: Governed metric definition for Hypercare Issue Burn-Down.

P20-KPI-060 - Stabilization SLO Attainment: Governed metric definition for Stabilization SLO Attainment.

P20-KPI-061 - Platform Availability: Governed metric definition for Platform Availability.

P20-KPI-062 - Platform Reliability Score: Governed metric definition for Platform Reliability Score.

P20-KPI-063 - Platform Security Score: Governed metric definition for Platform Security Score.

P20-KPI-064 - Platform Readiness Score: Governed metric definition for Platform Readiness Score.

P20-KPI-065 - Platform Digital Twin Freshness: Governed metric definition for Platform Digital Twin Freshness.

14. Engine Registry

P20-ENG-001 - Integration Contract Engine (Deterministic): Provides integration contract engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-002 - API Governance Engine (Deterministic): Provides api governance engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-003 - Schema Compatibility Engine (Deterministic): Provides schema compatibility engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-004 - Event Routing Engine (Deterministic): Provides event routing engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-005 - Message Idempotency Engine (Deterministic): Provides message idempotency engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-006 - Retry/Dead Letter Engine (Deterministic): Provides retry/dead letter engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-007 - Data Sync Engine (Deterministic): Provides data sync engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-008 - Reconciliation Engine (Deterministic): Provides reconciliation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-009 - Conflict Resolution Engine (Deterministic): Provides conflict resolution engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-010 - Identity Federation Engine (Deterministic): Provides identity federation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-011 - Authorization Policy Engine (Deterministic): Provides authorization policy engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-012 - Connector Health Engine (Deterministic): Provides connector health engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-013 - Environment Promotion Engine (Deterministic): Provides environment promotion engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-014 - Configuration Drift Engine (Deterministic): Provides configuration drift engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-015 - Feature Flag Engine (Deterministic): Provides feature flag engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-016 - Release Orchestration Engine (Deterministic): Provides release orchestration engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-017 - Artifact Integrity Engine (Deterministic): Provides artifact integrity engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-018 - Deployment Strategy Engine (Deterministic): Provides deployment strategy engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-019 - Rollback Engine (Deterministic): Provides rollback engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-020 - Infrastructure Policy Engine (Deterministic): Provides infrastructure policy engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-021 - Infrastructure Drift Engine (Deterministic): Provides infrastructure drift engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-022 - Schema Migration Engine (Deterministic): Provides schema migration engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-023 - Backup Verification Engine (Deterministic): Provides backup verification engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-024 - Restore Validation Engine (Deterministic): Provides restore validation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-025 - Threat Model Engine (Deterministic): Provides threat model engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-026 - Vulnerability Gate Engine (Deterministic): Provides vulnerability gate engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-027 - Dependency Risk Engine (Deterministic): Provides dependency risk engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-028 - Privacy Control Engine (Deterministic): Provides privacy control engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-029 - Tenant Isolation Engine (Deterministic): Provides tenant isolation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-030 - Observability Correlation Engine (Deterministic): Provides observability correlation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-031 - Alert Routing Engine (Deterministic): Provides alert routing engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-032 - SLI/SLO Engine (Deterministic): Provides sli/slo engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-033 - Error Budget Engine (Deterministic): Provides error budget engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-034 - Reliability Risk Engine (Deterministic): Provides reliability risk engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-035 - Performance Test Engine (Deterministic): Provides performance test engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-036 - Capacity Forecast Engine (Deterministic): Provides capacity forecast engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-037 - Autoscaling Recommendation Engine (Deterministic): Provides autoscaling recommendation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-038 - DR Readiness Engine (Deterministic): Provides dr readiness engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-039 - RTO/RPO Validation Engine (Deterministic): Provides rto/rpo validation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-040 - Test Orchestration Engine (Deterministic): Provides test orchestration engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-041 - Test Coverage Engine (Deterministic): Provides test coverage engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-042 - Regression Selection Engine (Deterministic): Provides regression selection engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-043 - Flaky Test Detection Engine (Deterministic): Provides flaky test detection engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-044 - Defect Triage Engine (Deterministic): Provides defect triage engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-045 - Requirements Traceability Engine (Deterministic): Provides requirements traceability engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-046 - Migration Validation Engine (Deterministic): Provides migration validation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-047 - Cutover Readiness Engine (Deterministic): Provides cutover readiness engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-048 - Production Readiness Engine (Deterministic): Provides production readiness engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-049 - Release Risk Engine (Deterministic): Provides release risk engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-050 - Incident Severity Engine (Deterministic): Provides incident severity engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-051 - Incident Correlation Engine (Deterministic): Provides incident correlation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-052 - RCA Evidence Engine (Deterministic): Provides rca evidence engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-053 - Service Level Engine (Deterministic): Provides service level engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-054 - Tenant Provisioning Engine (Deterministic): Provides tenant provisioning engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-055 - Entitlement Engine (Deterministic): Provides entitlement engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-056 - White-Label Configuration Engine (Deterministic): Provides white-label configuration engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-057 - Compliance Evidence Engine (Deterministic): Provides compliance evidence engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-058 - Access Review Engine (Deterministic): Provides access review engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-059 - AI Evaluation Engine (AI-assisted): Provides ai evaluation engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-060 - AI Guardrail Engine (AI-assisted): Provides ai guardrail engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-061 - AI Production Monitor Engine (AI-assisted): Provides ai production monitor engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-062 - AI Rollback Engine (AI-assisted): Provides ai rollback engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-063 - Hypercare Health Engine (AI-assisted): Provides hypercare health engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-064 - Stabilization Engine (AI-assisted): Provides stabilization engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-065 - Platform Control Tower Engine (AI-assisted): Provides platform control tower engine using governed environment, integration, security, test, operational and release evidence.

P20-ENG-066 - Platform Digital Twin Engine (AI-assisted): Provides platform digital twin engine using governed environment, integration, security, test, operational and release evidence.

15. Workflow Registry

P20-WF-001 - Integration Intake: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-002 - Integration Contract Design: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-003 - Connector Configuration: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-004 - API Publication: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-005 - Event Schema Publication: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-006 - Sync/Reconciliation: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-007 - Identity Federation Setup: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-008 - Environment Provisioning: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-009 - Configuration Promotion: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-010 - Feature Flag Promotion: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-011 - Secret/Certificate Rotation: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-012 - CI Build: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-013 - Automated Test Pipeline: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-014 - Artifact Signing: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-015 - Infrastructure Plan/Apply: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-016 - Schema Migration: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-017 - Deployment to Development: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-018 - Deployment to QA: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-019 - Deployment to Staging: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-020 - Production Deployment: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-021 - Canary/Phased Release: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-022 - Blue-Green Release: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-023 - Rollback: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-024 - Release Approval: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-025 - Change Record: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-026 - Post-Deployment Validation: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-027 - Observability Setup: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-028 - SLO/Error Budget Setup: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-029 - Performance Test: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-030 - Security Test: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-031 - Accessibility Test: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-032 - Disaster Recovery Test: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-033 - Backup Restore Test: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-034 - End-to-End Regression: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-035 - UAT Cycle: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-036 - Requirements Coverage Review: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-037 - Defect Triage: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-038 - Data Migration Assessment: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-039 - Migration Rehearsal: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-040 - Cutover Rehearsal: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-041 - Production Cutover: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-042 - Migration Reconciliation: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-043 - Production Readiness Review: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-044 - Go-Live Decision: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-045 - Tenant Provisioning: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-046 - White-Label Tenant Setup: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-047 - Incident Response: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-048 - Major Incident Communication: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-049 - Problem/RCA: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-050 - Corrective Action Tracking: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-051 - Compliance Evidence Collection: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-052 - Access Review: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-053 - AI Production Evaluation: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-054 - AI Guardrail Release: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-055 - AI Rollback: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-056 - Hypercare Command Center: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-057 - Stabilization Review: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-058 - Operational Handoff: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-059 - Platform Digital Twin Refresh: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

P20-WF-060 - Final Enterprise Acceptance & Lock: Initiate > Validate > Build/Test > Review > Approve > Deploy/Execute > Verify > Monitor > Close

16. Final Enterprise Acceptance & Lock Gate

Phase 20 remains review-ready until all required Product, Architecture, Engineering, QA, DevOps/SRE, Security/Privacy, Data Governance, Integration, AI Governance, Support/Service Management, Business Operations and UAT approvals are complete. After Phase 00-20 traceability reconciliation, blocking-gap closure and approval, the complete GES v1.0 Enterprise Specification may be locked as the governed implementation baseline.